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ORIGINAL RESEARCH

Strategic Alignment and Value Optimization: Unveiling the Critical Role of Project Portfolio Management for a Flexible Environment

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Abstract *Project portfolio management has emerged as a critical determinant for the success of both organizations and their constituent projects, and the evolving nature of a project portfolio and their objectives creates a necessity for flexibility within organizations. Despite the burgeoning interest among scholars and practitioners in project management criteria, there remains a paucity of empirical research delineating the mechanisms through which project portfolio management influences the value accrued by such portfolios. Employing a sample of project management professionals, this study utilized Structural Equation Modelling with mediation analysis and Necessary Condition Analysis to empirically evaluate a theoretical framework. This framework posits organizational strategy as a facilitative mechanism for the interrelation between flexible portfolio management and value creation and identifies the requisite and critical levels of portfolio selection criteria, portfolio management, and organizational strategy necessary for realizing project value. The findings contribute to a nuanced understanding of project value creation through portfolio management and underscore the strategic imperative of organizational strategy to enhance companies' adaptability and responsiveness. Additionally, the study delineates the essential roles of risk and change management, alongside portfolio management, from foundational levels of perceived value to advanced levels necessitated by impact criteria and resource management strategies.*

Keywords Dynamic adaptation · Flexible systems · Project portfolio management · Portfolio selection criteria · Value in projects

JEL Classification M1 · M11

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Introduction

The decision-making process for selecting optimal projects is crucial for efficient resource allocation in an interconnected business environment. Organizations face the challenge of choosing activities to undertake, balancing dynamic resources, capabilities, and goals (Morton et al., 2016). Project selection significantly contributes to organizational progress (Abbasianjahromi & Rajaie, 2012) and is a complex process influenced by evolving objectives (Chatterjee et al., 2018). Benaija (2015) notes that this phase involves comparing projects across changing dimensions to rank their suitability. Project portfolio management enables organizations to align agility with strategic flexibility, enhancing alignment with strategic goals under uncertainty (Dimple & Tripathi, 2024, Saeed et al., 2022).

According to PMI (2017), a project portfolio is a dynamic collection of projects, programmes, and operations managed to achieve strategic objectives. It emphasizes the interrelationship between projects, ensuring they collectively contribute to broader goals while maintaining adaptability to changing conditions. Unlike individual projects, which deliver specific outcomes, project

portfolios focus on alignment and flexibility within a systems paradigm (da Costa Filho et al., 2022). For instance, a technology firm may include product development, IT upgrades, and market expansion in its portfolio. These projects are dynamically managed to address market shifts and technological changes while achieving long-term goals of innovation, efficiency, and competitiveness, ensuring strategic alignment and sustainable value (Fuentes et al., 2019). This selection becomes even more challenging with the presence of project systematic interdependencies, where project characteristics change depending on the set of selected projects (Ganji et al., 2016). In a project-based organization, the proper selection of projects and resource allocation are critical issues where the complexity in the decision-making process increases as the number of projects grows (Chatterjee et al., 2018). In non-project-based organizations, the use of resources may go unnoticed, as projects are conducted alongside other activities. In this study, the investigation's scope was deliberately designed to encompass both types of organizations to enable a comprehensive analysis. The primary focus was concentrated on the portfolio management process, as it serves as a critical mechanism for aligning organizational resources with strategic objectives in a flexible business environment. Despite the growing emphasis on adaptability and resilience in modern organizations, limited research has explored how flexible systems can serve as a framework for enhancing portfolio management practices in project-based organizational models. While traditional approaches to portfolio management often prioritize standardization and control, they may lack the organizational agility needed to address the complexities of project selection criteria, ensure effective project portfolio management, and maximize portfolio value generation in rapidly changing and diverse operational environments.

This gap highlights the need for a deeper understanding of how flexibility in systems can be leveraged to optimize key processes such as decision-making, resource allocation, and prioritization across multiple projects. Specifically, there is limited insight into how flexible systems enable organizations to balance the competing demands of stability and innovation, while aligning project selection criteria and portfolio management practices with strategic objectives (Sushil, 2017). Moreover, the role of organizational strategy as a mediator between flexible systems and portfolio value generation remains largely unexplored, emphasizing the need for research that integrates strategic alignment into the framework of portfolio management.

By addressing this gap, the proposed research seeks to investigate the interplay between flexible systems, project selection criteria, project portfolio management practices, and portfolio value generation, with organizational strategy serving as a critical mediating factor. This nuanced perspective aims to uncover how organizations can dynamically tailor and refine their approaches to balance agility with control, align projects with strategic priorities, and achieve superior outcomes in complex and uncertain environments. This investigation is particularly relevant today, as adaptability and strategic alignment have become essential drivers of organizational success in increasingly volatile business contexts. The study determines the interplay between flexible systems and portfolio management practices, providing a nuanced perspective on how organizations can dynamically tailor and refine their approaches to achieve better outcomes, in particular project portfolio perceived value. The following statement summarizes the research question of this study: to what extent project portfolio selection criteria influence project portfolio management, and portfolio value generation in project-based organizations, mediated by organizational strategy, through the lens of flexible systems? This investigation is particularly relevant in today's complex and uncertain environments, where adaptability has become a key driver of organizational success.

The influence of project portfolio management on investment decisions highlights risks associated with poor management, including unnecessary investments, excessive risks, and lack of strategic alignment (Baptestone & Rabechini, 2018). Portfolio decisions are inherently challenging due to uncertain environments, incomplete information, and evolving systems (Benaija, 2015). Strategic alignment and leadership flexibility are critical to success, especially within the context of flexible systems management, which emphasizes adaptability and responsiveness to dynamic environments (Dhar et al., 2022). Poor portfolio management can result in suboptimal resource allocation and selection of projects that fail to deliver value to stakeholders (Chiang & Nunez, 2013). Value, understood as the subjective perception of project benefits, requires dynamic and adaptable management approaches (Cândido Da Silva et al., 2024; Martinsuo, 2020; Ruekert, 1992).

Despite its importance, the relationship between portfolio management and project value remains underexplored. Within flexible systems management, value evolves continuously with stakeholder perspectives, organizational priorities, and external conditions, necessitating agility and adaptability (Bizarrias et al., 2020; Tiwari & Suresha, 2021). Projects within portfolios contribute diverse forms of value, including strategic alignment, innovation, and social impact, which must be dynamically managed to remain relevant to shifting organizational objectives (De la Gala-Velásquez et al., 2023). Flexibility is also central to benefit realization management, ensuring that the value generated aligns with both internal and external changes (Serra, 2016). This study addresses gaps in understanding the role of organizational strategy, framed within flexible

systems management, as a mediator between project portfolio criteria and value generation.

Organizational strategy within a flexible systems paradigm acts as a balancing mechanism, aligning portfolio management with dynamic project outcomes (Vishvakarma et al., 2021). Strategic alignment ensures that project selection criteria are tailored to organizational goals and adjusted to accommodate shifting priorities (Ansari et al., 2015; Ul Musawir et al., 2017). Studies emphasize that strategic alignment, when embedded within a flexible systems approach, enhances customer satisfaction and organizational adaptability, contributing to sustainable value creation (Bentzen et al., 2011; Ika & Pinto, 2022). Selection criteria must encompass technical, financial, and managerial feasibility, alongside risk considerations, to optimize portfolio value in dynamic environments (Costantino et al., 2015).

This study integrates the Resource-Based View (RBV) (Barney et al., 2021) with Dynamic Capabilities Theory (Teece, 2007) and flexible systems management to explore how strategic alignment enhances portfolio management. RBV identifies competitive advantages through unique resources, while Dynamic Capabilities Theory and flexible systems management emphasize agility in adapting resources and competencies to evolving contexts. Strategic alignment, as a dynamic capability, ensures that projects within portfolios are dynamically managed to optimize value creation and resource utilization. Methodologically, data were collected from project portfolio management professionals. Structural Equation Modelling (SEM) (Hair et al., 2022) revealed direct and mediated relationships (Hayes, 2022), while Necessary Condition Analysis (Dul et al., 2023) identified critical thresholds for value generation. This study advances the field by integrating flexible systems management, highlighting its role in fostering adaptability, optimizing strategic alignment, and enabling continuous value creation amidst complex and changing environments.

Theoretical foundations

Value in project management, organizations resources, and capabilities

The study of project portfolio management (PPM) often emphasizes financial outcomes while overlooking broader aspects of value management (Martinsuo & Killen, 2014). Traditional metrics like return on investment dominate, yet value is increasingly recognized as a subjective perception shaped by project outcomes and benefits (Browning, 2014; Cândido da Silva et al., 2024; Martinsuo, 2020). Effective PPM aims to align projects with strategic objectives, optimizing value while balancing risks and resources (Chatterjee et al., 2018; Tavana et al., 2019). Combining Resource-Based View (RBV) and Dynamic Capabilities Theory provides a robust framework for understanding value creation in PPM (Almarri & Gardiner, 2014). RBV focuses on leveraging valuable, rare, inimitable, and non-substitutable resources for competitive advantage, while Dynamic Capabilities Theory emphasizes adapting competencies to changing environments (Barney et al., 2021).

In PPM, these theories highlight the importance of aligning project selection with strategic goals and managing interdependencies effectively (Martinsuo & Geraldi, 2020). Flexibility and adaptability are essential for refining capabilities to address environmental changes and sustain competitive advantage. Beyond financial returns, PPM integrates strategic, ecological, and social criteria to enhance long-term value (Martinsuo, 2020). This comprehensive approach ensures project portfolios align with organizational missions, fostering sustainable value creation (Martinsuo & Killen, 2014).

Project portfolio management and value creation

The foundational principles of portfolio management began with Markowitz's Modern Portfolio Theory, which significantly influenced quantitative approaches in finance and investment (Markowitz, 1952). This framework parallels project portfolio management (PPM), where evaluating project value is essential for effective selection and strategic alignment (Gertzen et al., 2022).

PPM is central to organizational decision-making, guiding prioritization amidst risks, costs, and dynamic interdependencies (Chatterjee et al., 2018; Kaiser et al., 2019). Beyond operational concerns like resource allocation and risk management, PPM represents a strategic endeavour focused on generating long-term value (Martinsuo, 2013). However, the mechanisms by which PPM achieves this value, particularly at the portfolio level, remain underexplored (Cândido Da Silva et al., 2024).

PPM extends beyond managing individual projects, emphasizing how portfolios collectively drive innovation, competitive advantage, and strategic growth (Chiang & Nunez, 2013). Traditional perspectives often neglect the iterative nature of value creation, focusing instead on efficiency or project timelines. True success, however, involves post-implementation benefits, innovation capacity, and alignment with stakeholder perceptions of value (Aghajani et al., 2023).

Research highlights gaps in understanding how PPM balances project criteria, stakeholder needs, and market conditions to sustain value (Cable et al., 2004; Malanchuk et al., 2023). Addressing these gaps could better optimize

resource utilization and strategic outcomes (Tavana et al., 2019).

+H1. Project portfolio management positively influences the generation of portfolio value.

Organizational strategy is a cornerstone of project portfolio management (PPM), linking strategic intent to portfolio value. This alignment ensures project investments are optimized for long-term organizational goals and return on investment (ROI) (Alzoubi & Yanamandra, 2020). By aligning portfolios strategically, organizations achieve sustainable competitive advantages, resource efficiency, and market resilience (Abbas, 2020). In dynamic environments, this adaptability is crucial for responding swiftly to market changes (Sawaean & Ali, 2020).

Strategic oversight enhances PPM decision-making by prioritizing projects based on organizational objectives, thereby maximizing portfolio value (Orazalin, 2020). Effective integration of strategy fosters innovation and resilience, enabling firms to adapt to complexity while aligning projects with growth opportunities (Kock et al., 2020). Yao et al. (2023) found that strategic alignment improves project outcomes and ROI, while Petro et al. (2020) emphasized its role in achieving long-term goals. Challenges arise within the flexible systems paradigm, which requires balancing adaptability and strategic alignment. While strategy targets long-term objectives, PPM focuses on dynamic resource allocation and value optimization (Srivannaboon & Milosevic, 2006). Addressing this gap involves frameworks accommodating rapid market changes while maintaining strategic coherence (Chowdhury et al., 2024). Ultimately, organizational strategy ensures alignment, optimizes ROI, and enhances PPM effectiveness (Abbas, 2020; Alzoubi & Yanamandra, 2020; Kock et al., 2020; Orazalin, 2020; Petro et al., 2020; Sawaean & Ali, 2020; Yao et al., 2023).

+H1'. Organizational strategy mediates the relationship between project portfolio management and project portfolio value.

Project selection involves evaluating changing criteria such as cost, risk, impacts, and deadlines, critical for portfolio prioritization (Chatterjee et al., 2018). The lack of consensus on selection criteria highlights the need for flexibility in adapting to distinct scenarios (Costantino et al., 2015; Dutra et al., 2014; Yousefi et al., 2018). Cost and risk are primary factors, aligning with investor priorities under uncertainty (Kaiser et al., 2019; Srizongkhram et al., 2020). Financial considerations focus on maximizing returns through efficient resource allocation (Frej et al., 2021), while robust risk analysis is essential for managing uncertainties (Mavrotas & Pechak, 2013). The analysis of extant literature on project portfolio criteria is shown in Table 1.

The absence of strategic financial criteria in project portfolio management often leads to a diminished perception of project value. This perception is vital for the success and sustainability of ventures as it reflects the project's ability to meet stakeholders' expectations, generate return on investment, and adapt to market conditions. Strategic financial criteria ensure that projects align with the organization's overall financial objectives, optimizing resource use and maximizing returns. Without solid financial planning, projects risk exceeding budgets and being perceived as poorly managed (Albano et al., 2019; Dreyer et al., 2022).

+H1nc1↑. The Strategic Financial Criterion is a necessary condition for portfolio value.

Risk and change management are crucial for navigating the uncertainties of project lifecycles, enabling organizations to anticipate and mitigate challenges effectively. This management aspect goes beyond preventing negative outcomes; it enhances an organization's adaptability and resilience in volatile environments. Effective change management ensures smooth implementation of necessary adaptations, enhancing stakeholder trust and support for projects (Abbasianjahromi & Rajaie, 2012; Frej et al., 2021; Obondi, 2022).

+H1nc2↑. The criterion of risk and change management is a necessary condition for portfolio value.

Resource and impact management involve the strategic allocation and optimization of human, financial, and material resources to achieve project objectives. Efficient management impacts project quality, timing, and costs. Inadequate resource management can lead to delays, compromised project outcomes, and negative stakeholder perceptions, ultimately affecting the organization's reputation and viability (Bolívar et al., 2012; Ganji et al., 2016).

+H1nc3↑. The criterion of resource management and impact is a necessary condition for the value of the portfolio.

Innovation and technological development are imperative for maintaining competitiveness in a globalized market. These elements involve the introduction of new processes, technologies, and management methods that increase project efficiency and effectiveness. Innovations anticipate future market demands, ensuring project relevance over time, and fostering sustainable practices, which enhance the corporate image and open new market opportunities (Azimi et al., 2019; Costantino et al., 2015). Innovation is particularly important when companies deal

Table 1 Project selection criteria by group

Studies	Main project selection criteria	Groups
Abbasianjahromi and Rajaie (2012)	Strategic alignment	Strategic and Financial Criteria
Albano et al., (2019)	Financial viability	
Chiang and Nunez (2013)	Impact on the organization's image	
Dreyer et al., (2022)		
García-Melón et al., (2015), Frej et al., (2021)		
Montajabiha et al., (2017)		
Song et al., (2021)		
Abbasianjahromi and Rajaie (2012), Bolívar et al., (2012)	Level of risk and uncertainty	Risk and Change management criteria
Dutra et al., (2014)	Contribution to organizational learning	
Frej et al., (2021)	Adaptation to changing market conditions	
Montajabiha et al., (2017)		
Song et al., (2021)		
Abbasianjahromi and Rajaie (2012)	Availability of resources	Resource Management and Impact Criteria
Benaija (2015)	Environmental impact	
Bolívar et al., (2012)	Social impact	
Ganji et al., (2016)	Compatibility with organizational culture	
Martinsuo (2013)		
Azimi et al., (2019)	Contribution to technological development	Innovation and Technological Development Criteria
Costantino et al., (2015) Dreyer et al., (2022)		
Bolívar et al., (2012)	Contribution to innovation development	

with risky and flexible environments and organizations challenges (Tiwari & Suresha, 2021).

+H1nc4↑. The criterion of innovation and technological development is a necessary condition for the value of the portfolio.

Project portfolio management focuses on the selection and prioritization of projects to reinforce strategic objectives. This management practice is not just administrative but strategic, aligning projects with corporate goals to maximize overall success. Effective portfolio management ensures optimal resource allocation and identifies synergies among projects, preventing resource dispersion and ensuring strategic alignment (Martinsuo, 2013).

+H1nc5↑. Project portfolio management is a necessary condition for portfolio value.

Organizational strategy in project portfolio management

Project portfolio management is intricately linked to organizational strategy, which acts as a foundational framework for aligning project management processes with strategic objectives (Cooper et al., 2001; Kaiser et al., 2019). Haniff and Fernie (2008) describe organizational strategy as a plan that an organization adopts to meet its goals, influenced by factors such as stakeholder presence and the need for strategic alignment in projects. It is widely acknowledged that organizational strategy facilitates the achievement of strategic objectives (Green et al., 2006; López-Cabarcos et al., 2015).

Canbaz and Marle (2016) discuss the critical role of organizational strategy in ensuring that portfolio management reflects the strategic priorities of the organization, aligning resource allocation and project execution with overarching strategic goals. Similarly, Chiang and Nunez (2013) and Gerow et al. (2015) emphasize the importance of strategic alignment between organizational strategy and portfolio management to enhance the organization's capability to realize strategic objectives and optimize value creation.

+H2. Project portfolio management positively influences the organization's strategy.

Organizational strategy is essential for creating and perceiving project value, providing the roadmap for project selection, prioritization, and execution, thus ensuring projects align with long-term organizational objectives (Bai et al., 2022; Martinsuo et al., 2022). It acts as a

communication tool, conveying the company's purpose and direction to stakeholders, enhancing project value perception by linking projects to organizational goals.

Strategy is the essence of an organization's direction and purpose. Without a coherent organizational strategy, projects risk becoming disconnected from corporate goals, reducing their perceived value. It is fundamental for performance evaluation and success measurement, creating a value system by which project success is measured. The absence of a clear organizational strategy compromises not only the generation of value but also the perception and recognition of value in project efforts, driven by critical factors for project success and positive outcomes (Lamprou & Vagiona, 2022).

H2nc↑. Organizational strategy is a necessary condition for portfolio value.

This synthesis highlights the indispensable role of organizational strategy in project portfolio management, underlining its significance in aligning projects with the strategic objectives of an organization and enhancing their value perception and contribution.

+H3. The organization's strategy positively influences the generation of portfolio value.

Taken together, these hypotheses can be observed in the proposed model, seen in Fig. 1.

Method

To better attain the objective of the study and answer the research question, we systematically employed a three-step methodological approach, with an interconnected range of techniques. Due to the lack of consensus regarding measurement instruments for project portfolio evaluation, we conducted an in-depth analysis of the extant literature to identify key criteria. Based on these findings, we developed composite, or formative constructs that capture organizational attributes rather than latent variables at the individual level. After this procedure, we first conducted an exploratory step of model adjustment. Secondly, these composites were incorporated into a research design to elucidate the phenomenon of project portfolio value, conceptualized as being mediated by organizational strategy, analysed through Structural Equation Modelling. Subsequently, we applied Necessary Condition Analysis (NCA) to complement the study. This methodological approach enabled us to establish the requisite threshold levels of the predictor composites necessary for the realization of project portfolio value. By integrating these analytical techniques, we sought to provide a more comprehensive understanding of the phenomenon and the interrelations among its constituent elements.

Measures

A Systematic Literature Review (SLR) was conducted to investigate project portfolio selection criteria, following Pollock and Berge's (2018) protocol. After filtering 57 studies from Scopus and Web of Science, relevant criteria were identified and grouped into dimensions by experts in project portfolio management. These criteria formed the basis for developing formative constructs (Henseler et al., 2020), reflecting organizational indices for project portfolio evaluation (Blomquist et al., 2016; Xu et al., 2019).

The constructs included strategic and financial criteria, risk and change management, resource and impact management, and criteria for technological development and innovation. PPM was defined as a centralized process for

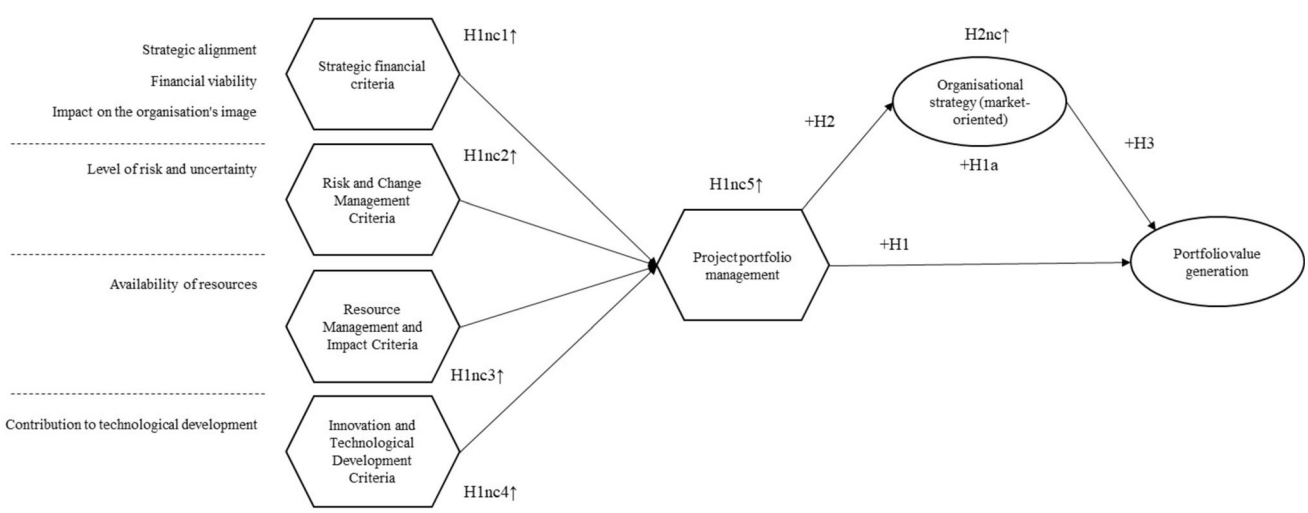

Fig. 1 Conceptual model

selecting and managing portfolios aligned with strategic objectives, optimizing resources, managing risks, and maximizing portfolio value (Abbasianjahromi & Rajaie, 2012; Dreyer et al., 2022; Martinsuo, 2013). Strategic criteria prioritize alignment with long-term goals (Albano et al., 2019), while financial criteria focus on economic viability and ROI (García-Melón et al., 2015). Risk and change management address uncertainties and adaptability (Frej et al., 2021). Resource and impact management ensure efficient resource use and stakeholder benefits (Martinsuo, 2013). Technological criteria foster innovation and competitive advantage (Azimi et al., 2019).

Data were analysed using PLS-SEM for predictive modelling (Hair et al., 2022) and Necessary Condition Analysis (NCA) to identify critical antecedent levels for portfolio value (Dul et al., 2023). This complementary approach combines predictive insights with essential condition identification, offering a nuanced understanding of portfolio value generation. NCA highlights the non-compensatory relationships between variables, emphasizing the critical thresholds required for value occurrence (Richter et al., 2020, 2023). These methods provide a robust framework for examining how strategic alignment and selection criteria drive value in project portfolios.

Data Collection

The research instrument underwent a preliminary review by two project management experts, who suggested modifications for improved semantic clarity of the items, which were subsequently implemented. To validate the research model, a survey was conducted targeting organizations engaged in PPM. The measurement tool was distributed electronically to potential respondents via business platforms and project management interest groups, ensuring a comprehensive and suitably qualified sample profile for the study's objectives. Participants were informed about the survey on PPM, and those who consented to participate and confirmed their involvement in such projects were directed to the online data collection platform.

The sample for this study comprised participants engaged in project management activities, primarily focused on project portfolio management. Sample size was planned to exceed a minimum of 154 valid responses, based on Hair et al., (2022, p. 26) following the p-value and level of significance (inverse square root method) expressed in the equation $\left(n = \left(\frac{2.486}{\text{path min}}\right)^2\right.$, for a pathmin = 0.2, and power level of 80%). Following G.Power, and the same criteria of power, and 2 predictors, the minimum sample size was 107 (Bartlett, 2019).

A total of 198 responses were collected via survey questionnaire, of which 10.2% were associated with agile projects, 44.0% with hybrid projects, 44.9% with traditional projects. Among the respondents, 27% were project managers, 23% were project portfolio managers, 18.9% were project analysts, and 17.5% were project coordinators. Regarding educational background, 57.1% had degrees in engineering, approximately 19.4% in exact and earth sciences, and 8.2% in applied social sciences. A notable aspect of the sample is the practitioners' experience levels, with 31.6% having 11 to 20 years of experience, 27.6% having 1 to 5 years, 21.4% having 6 to 10 years, 12.2% having 21 to 30 years, and 7.1% having more than 30 years of experience.

Results

The results are presented in alignment with the three-step methodological approach outlined in Method section, ensuring a clearer and more structured understanding of the study's findings.

Model adjustment

The structural model was assessed for reliability, convergent validity, and discriminant validity (Hair et al., 2022) before testing hypotheses. Reflective and formative measures were included to account for variable diversity. To mitigate common method bias, Podsakoff et al. (2003) guidelines were applied, including randomization during data collection to prevent response bias, Harman's single-factor test (variance explained: 12.62%, KMO = 0.874, $\chi^2 = 1910.95$, $p < 0.001$), and variance inflation factor (VIF) analysis (Chin et al., 2013; Kock, 2015), with no critical levels detected. Reliability was confirmed via AVE, R^2, adjusted R^2, and Cronbach's Alpha, with convergent and discriminant validity verified using HTMT and Fornell–Larcker criteria. Taken together, these indicators proved adequate, as observed in Table 2.

These initial results of the model's fit proved to be appropriate and allowed the test of the hypotheses proposed in the study to be evaluated.

Hypothesis testing

The direct hypotheses of the proposed model were analysed using Structural Equation Modelling based on a variance matrix and partial least squares, using Smart PLS 4 software. Table 3 shows the initial results of the hypothesis testing, confirming the positive influence of the portfolio management criteria on portfolio management and on the organization's strategy, as well as the positive influence of the organization's strategy on the value of the project portfolio. In addition, the conditional role of the organization's strategy in enabling the influence of

Table 2 Model adjustment

Indicators	Reliability						Heterotrait monotrait			Fornell-Larcker		
Constructs	Cronbach's Alpha	CR $_{(rho_a)}$	CR $_{(rho_c)}$	AVE	R^2	R^2_{adj}	1	2	3	1	2	3
1. Organizational strategy	0,832	0.834	0.881	0.60	0.57	0.565	–	–	–	0.77	–	–
2. Project portfolio management	0,877	0.891	0.901	–	0.999	0.991	0.867	–	–	0.75	0.70	
3. Portfolio value generation	0,905	0.911	0.922	0.57	0.55	0.538	0.735	0.804	–	0.64	0.73	0.76

Table 3 Hypothesis testing

Hs	Structural path	Γ	Γ_{mean}	*S*	t	p	Status
H1	Project Portfolio Management → Portfolio Value	0.57	0.57	0.11	5.571	0.001	Supported
H2	Project Portfolio Management → Organizational Strategy	0.75	0.76	0.04	18.311	0.001	Supported
H3	Organizational Strategy → Portfolio Value	0.21	0.21	0.12	1.826	0.068	Supported
H1'	Project Portfolio Management → Organizational Strategy → Portfolio Value	0.16	0.16	0.09	1.835	0.067	Supported
–	Strategic Financial Criteria → Project Portfolio Management	0.16	0.16	0.03	6.053	0.001	–
–	Risk and Change Management Criteria → Project Portfolio Management	0.36	0.36	0.03	12.100	0.001	–
–	Resource Management Criteria Impact → Project Portfolio Management	0.38	0.38	0.03	12.790	0.001	–
–	Innovation and Technological Development Criteria → Project Portfolio Management	0.29	0.31	0.03	11.090	0.001	–

portfolio management on portfolio value was observed, confirming the study's hypotheses.

The values presented in the analysis offer insights into the estimates of path coefficients (Γ), complemented by various model fit measures such as $\Gamma_{_Average}$ and S. The inclusion of t-tests and p-values for construct relationships enhances understanding of interactions. For exploratory purposes, a 10% significance level was used for hypotheses H3 and H1' (Kline, 2013), allowing identification of patterns without overly restrictive thresholds. Kline emphasizes that validation should consider study design, sample characteristics, confidence intervals, and effect sizes. The average effect size of 1.27 with narrow confidence intervals confirms robust findings. Necessary Condition Analysis (NCA) further identifies critical levels of antecedents influencing portfolio value. This approach highlights essential conditions for portfolio value realization, advancing insights into strategic alignment and effective portfolio management practices.

Necessary conditions analysis.

The selection and prioritization of initiatives are crucial for aligning projects with strategic objectives and maximizing ROI. However, applying project selection criteria alone does not ensure value generation within the portfolio. This research employs Necessary Condition Analysis (Dul et al., 2023) to identify the extent, context, and critical levels at which selection criteria influence value generation. Using R Studio, interactions among selection criteria are analysed to determine necessary conditions for portfolio value perception across contexts. Results, detailed in Table 4, utilize the CR_FDH method while comparing other techniques. This nuanced analysis provides actionable insights for project managers and strategists, emphasizing sustainable value creation and competitive advantage in dynamic markets.

The analysis of the data indicates that the factors "Risk and Change Management Criterion" and "Project Portfolio Management" exhibit substantial predictive coefficients, with values of 0.33 and 0.27, respectively. These findings underscore the significance of these criteria in enhancing the value derived from project portfolio management. Additionally, the magnitude of the effect in the necessary condition analyses reveals the limitation of the outcome variable imposed by the predictor variable, comparable to other effect size metrics such as correlation coefficients or R^2 in structural equation models. The adequacy of the model is corroborated by the results. This analysis details how different project selection criteria contribute at various levels to the realization of specific values in the portfolio.

For example, to achieve an initial value of 20% in Project Portfolio Value, a minimum performance of 6.0% in the "Risk and Change Management" criterion is necessary. To achieve a 30% increase in Value Generation, a performance of 14.7 in "Risk and Change Management"

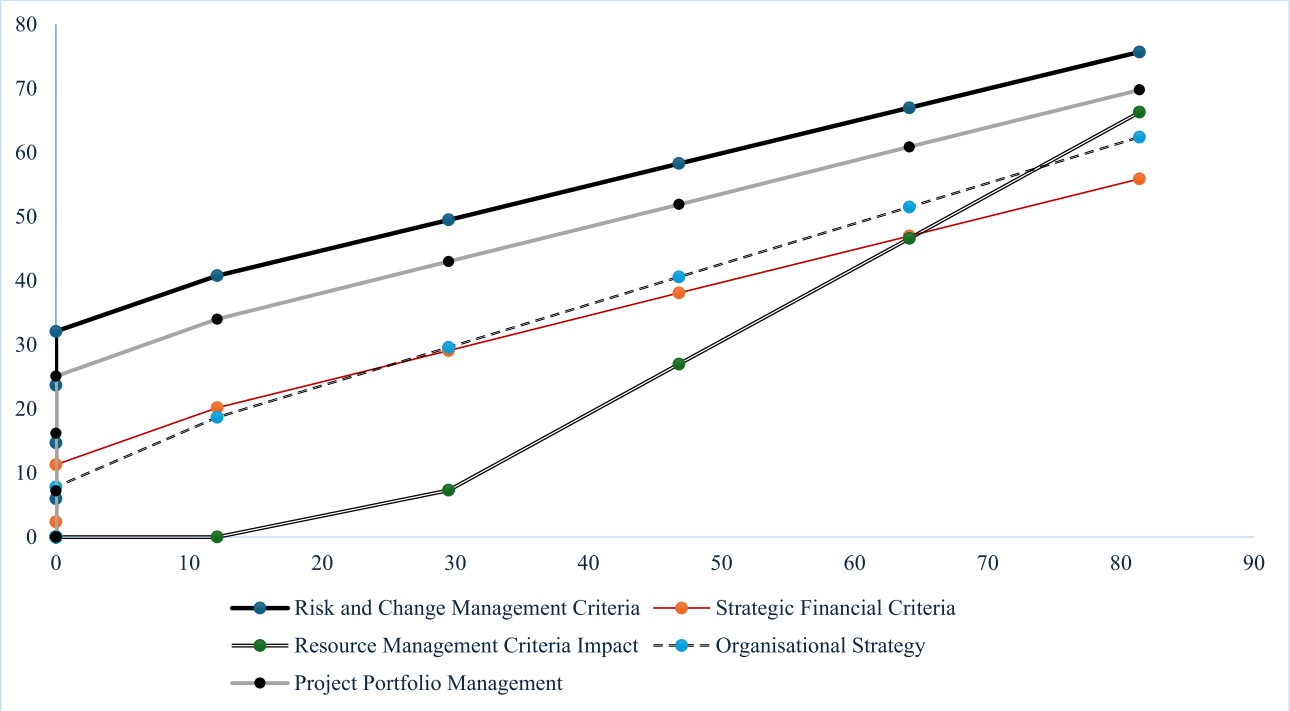

Fig. 2 Evolution of necessary conditions

and 7.2 in "Project Portfolio Management" is required. This progression continues analogously for higher levels of portfolio valuation. Figure 2 presents the development of the role of portfolio value antecedents as the project evolves and highlights the importance of risk management and portfolio management criteria. However, it points out the growing importance of resource management and impact criteria as the value of the portfolio intensifies, and an interesting decline in financial criteria.

Discussion

This study developed and evaluated a comprehensive framework to integrate project selection criteria with portfolio management processes, enhancing project portfolio value through organizational strategy mediation. Project selection criteria were categorized into four groups explicitly linked to value generation, contingent upon strategic alignment. This approach provides nuanced insights into how strategic considerations shape portfolio management decisions and foster value creation.

The empirical results supported all tested hypotheses, demonstrating the positive impact of effective project portfolio management (PPM) on portfolio value (H1: $\Gamma = 0.566$, $t = 5.573$, $p < 0.001$). These findings align with prior studies highlighting the significance of PPM in achieving project success and optimizing portfolio value (Flores, 2020; Petro & Gardiner, 2015). Unlike traditional project success measures (e.g., time, scope, quality), portfolio value encompasses strategic alignment, stakeholder satisfaction, and long-term organizational goals (Lamprou & Vagiona, 2022). Portfolio value is dynamic, influenced by shifting priorities and diverse stakeholder perspectives (Martinsuo, 2020). For instance, stakeholders may emphasize innovation, risk mitigation, or social impact. This diversity necessitates flexible management approaches to adapt to changing internal and external environments. A rigid PPM framework risks obsolescence, while iterative value assessments, resource reallocation, and stakeholder engagement ensure sustained alignment with strategic objectives (Voss & Kock, 2013).

A statistically significant relationship between PPM and organizational strategy was observed (H2: $\Gamma = 0.754$, $t = 18.296$, $p < 0.001$), emphasizing the importance of strategic alignment for portfolio performance. This confirms previous findings linking PPM to organizational strategy (Vishvakarma et al., 2021). Portfolio management serves as a bridge between strategic planning and project execution, ensuring resources are allocated to achieve strategic goals (Bucero, 2017; Petro & Gardiner, 2015;). Additional hypotheses (H3: $\Gamma = 0.213$, $p = 0.068$; H4: $\Gamma = 0.16$, $p = 0.067$) reinforced the mediating role of organizational strategy in linking PPM and portfolio value. This strategic framework aligns project selection and management processes with broader objectives, enhancing

effectiveness and value realization. The findings build on traditional frameworks (Kopmann et al., 2017; Meskendahl, 2010) by highlighting the direct impact of strategic alignment on portfolio value, a relatively underexplored outcome.

Kaczorowska et al. (2019) emphasize the subprocesses of identifying, categorizing, evaluating, and balancing portfolios as essential for optimizing resources and enabling economically justified project choices. These results advance existing knowledge by proposing that project portfolio value, rather than project success alone, serves as a key outcome of strategic alignment.

The study explored critical conditions for value creation within project portfolios. It identified risk and change management as pivotal for increasing perceived portfolio value by 20%. Achieving a 30% value increase required excellence in both risk management and overall PPM. These findings underscore the strategic importance of managing risks and changes effectively to optimize value (Browning, 2014; Willumsen et al., 2019). Willumsen et al. (2019) highlight the subjective dimensions of value creation in risk management, emphasizing alignment with stakeholder expectations. Browning (2014) complements this with a quantitative framework for balancing risk and opportunities, operationalizing their interplay to enhance value. Together, these perspectives align with the study's findings, demonstrating how contextualized risk management strategies support portfolio value optimization.

Strategic alignment significantly enhances PPM effectiveness, aligning project selection with organizational objectives. Bucero (2017) and Tiwari and Suresha (2021) underscore the role of portfolio management in operationalizing strategy and driving innovation. A strategic, integrative approach ensures projects complement each other, optimizing portfolio outcomes. This approach emphasizes not just immediate success but also long-term benefits such as sustainability, innovation, and competitive advantage. For instance, Kaczorowska et al. (2019) highlight the importance of balancing strategic relevance with risk analysis to enhance organizational outcomes. Similarly, strategic portfolio management requires systematic selection processes that consider uncertainty and adaptability (Lin & Hsieh, 2004; Benaija & Kjiri, 2014). The study highlights a shift away from traditional financial metrics in project evaluation. Financial criteria, while foundational, often fail to capture the broader contributions of modern projects, particularly in dynamic environments (Kock et al., 2020; Martinsuo, 2020). As projects progress, initial financial assessments must be supplemented with broader evaluative criteria, encompassing strategic, operational, and societal impacts.

Projects critical to long-term goals or risk mitigation might be undervalued if assessed solely on financial outcomes. A comprehensive approach to value assessment aligns project evaluation with strategic objectives, enhancing decision-making and ensuring projects deliver value beyond immediate financial gains (Serra & Kunc, 2015). By incorporating non-financial metrics, organizations can better evaluate projects' contributions to sustainability and competitive positioning.

This study contributes unique insights into how strategic alignment and PPM enhance portfolio value. It underscores the importance of integrating selection criteria with management processes to align portfolios with organizational objectives. Effective PPM depends on structural alignment, moving beyond project selection to strategy implementation (Kaiser et al., 2019). Implementing advanced decision support systems, such as fuzzy set theory, can further support strategic decision-making under uncertainty (Lin & Hsieh, 2004). Aligning project selection with organizational strategy also enhances portfolio value by emphasizing effective risk and change management (Benaija & Kjiri, 2015). The observed emphasis on broader value measures reflects the evolving view of project management as a tool for strategic alignment, fostering adaptability in changing markets (Green et al., 2006; López-Cabarcos et al., 2015). By expanding evaluation criteria to include non-financial metrics, organizations can better align project outcomes with strategic goals, driving long-term value creation and sustainability. This research highlights the critical role of strategic alignment in enhancing project portfolio value. By integrating project selection criteria with portfolio management processes and emphasizing flexible, iterative approaches, organizations can optimize resources, mitigate risks, and adapt to dynamic environments. This strategic, integrative approach not only ensures alignment with organizational objectives but also advances long-term competitiveness and adaptability.

The findings advance existing knowledge by linking organizational strategy directly to portfolio value, emphasizing the importance of managing risks and changes effectively. Future research should continue exploring these dynamics, particularly within diverse organizational contexts, to further refine the framework for value generation in PPM. Through its systematic approach, this study provides actionable insights for project managers and strategists, underscoring the importance of strategic integration in achieving sustainable portfolio success.

Project portfolio value management and flexible systems

This study developed a comprehensive framework to integrate project selection criteria with portfolio management processes, enhancing project portfolio value through organizational strategy mediation. Selection criteria were

categorized into four groups explicitly linked to value generation, emphasizing strategic alignment's influence on portfolio decisions and value creation.

Empirical results supported all tested hypotheses, demonstrating the positive impact of effective PPM on portfolio value (H1: $\Gamma = 0.566$, $t = 5.573$, $p < 0.001$). These findings align with prior studies emphasizing PPM's role in achieving project success and optimizing value (Flores, 2020; Petro & Gardiner, 2015). Unlike traditional success measures (time, scope, quality), portfolio value reflects strategic alignment, stakeholder satisfaction, and long-term goals (Lamprou & Vagiona, 2022). Portfolio value's dynamic nature, shaped by shifting priorities and stakeholder perspectives, necessitates flexible management approaches (Martinsuo, 2020). Iterative assessments, resource reallocation, and stakeholder engagement ensure sustained alignment with strategic objectives (Voss & Kock, 2013).

The study confirmed a significant relationship between PPM and organizational strategy (H2: $\Gamma = 0.754$, $t = 18.296$, $p < 0.001$), highlighting strategic alignment's role in portfolio performance. PPM bridges strategic planning and execution, ensuring resources align with strategic goals (Bucero, 2017; Petro & Gardiner, 2015). Hypotheses H3 ($\Gamma = 0.213$, $p = 0.068$) and H4 ($\Gamma = 0.16$, $p = 0.067$) reinforced organizational strategy's mediating role in linking PPM to portfolio value. This strategic framework aligns project selection with broader objectives, enhancing effectiveness and value realization. The findings build on frameworks by Meskendahl (2010) and Kopmann et al. (2017), emphasizing portfolio value as an underexplored outcome.

Kaczorowska et al. (2019) emphasize processes such as identifying, categorizing, evaluating, and balancing portfolios as critical for resource optimization and economically justified project choices. These findings advance knowledge by proposing that portfolio value, not just project success, reflects strategic alignment's benefits. The study identified risk and change management as pivotal for value creation. Excellence in risk management alone increased perceived portfolio value by 20%; combining it with effective PPM raised value by 30%. These findings underscore the strategic importance of managing risks and changes to optimize value (Browning, 2014; Willumsen et al., 2019). Willumsen et al. (2019) highlight value creation's subjective dimensions in risk management, emphasizing alignment with stakeholder expectations. Browning (2014) provides a complementary quantitative framework, operationalizing risk and opportunity management to enhance value. These perspectives align with this study's findings, illustrating how contextualized strategies support portfolio value optimization.

Strategic alignment significantly enhances PPM's effectiveness, aligning project selection with organizational objectives. Bucero (2017) and Tiwari and Suresha (2021) underscore PPM's role in operationalizing strategy and fostering innovation. A strategic, integrative approach ensures projects complement each other, optimizing outcomes.

This approach extends beyond immediate success to long-term benefits such as sustainability, innovation, and competitive advantage. Kaczorowska et al. (2019) emphasize balancing strategic relevance with risk analysis to enhance organizational outcomes. Similarly, Lin and Hsieh (2004) and Benaija & Kjiri (2015) advocate systematic selection processes that accommodate uncertainty and adaptability. The study highlights a shift from traditional financial metrics in project evaluation. Financial criteria, while foundational, fail to capture broader contributions, especially in dynamic environments (Kock et al., 2020; Martinsuo, 2020). As projects progress, initial financial assessments must expand to include strategic, operational, and societal impacts.

Projects critical to long-term goals or risk mitigation may be undervalued when assessed solely on financial outcomes. Incorporating non-financial metrics enables organizations to evaluate projects more holistically, aligning outcomes with strategic goals and ensuring value beyond immediate gains (Serra & Kunc, 2015). This study provides unique insights into how strategic alignment and PPM enhance portfolio value. Integrating selection criteria with management processes ensures alignment with organizational objectives. Effective PPM requires structural alignment, moving beyond project selection to strategy implementation (Kaiser et al., 2019).

Advanced decision support systems, like fuzzy set theory, support strategic decision-making under uncertainty (Lin & Hsieh, 2004). Aligning project selection with organizational strategy enhances portfolio value by emphasizing effective risk and change management (Benaija & Kjiri, 2015). The shift to broader value measures reflects project management's evolving role as a strategic alignment tool, fostering adaptability in changing markets (Green et al., 2006; López-Cabarcos et al., 2015). Expanding evaluation criteria to include non-financial metrics better aligns project outcomes with strategic goals, driving long-term value and sustainability.

This study emphasizes strategic alignment's critical role in enhancing project portfolio value. By integrating project selection criteria with portfolio management processes and emphasizing flexibility, organizations can optimize resources, manage risks, and adapt to dynamic environments. This strategic, integrative approach ensures alignment with objectives while advancing long-term competitiveness. The findings link organizational strategy

directly to portfolio value, highlighting the importance of managing risks and changes. Future research should explore these dynamics further, refining frameworks for value generation in diverse contexts. This systematic approach offers actionable insights for project managers and strategists, supporting sustainable portfolio success.

Practical Implications

This study emphasizes the interconnection between PPM and organizational strategy, highlighting strategic alignment, risk management, and adaptability as key to value generation. Flexible PPM systems allow organizations to adjust portfolios in response to dynamic conditions. For example, automotive companies might balance R&D for electric vehicles with initiatives to enhance production efficiency, ensuring alignment with evolving goals and market demands. Strategic decision-making in PPM requires adaptable approaches. In pharmaceuticals, firms prioritize high-potential drug development projects while reallocating resources when regulatory or market shifts occur. Similarly, financial institutions use PPM to navigate economic downturns, shifting focus from high-risk ventures to efficiency-driven projects, mitigating disruptions while safeguarding strategic objectives.

Risk management emerges as a core determinant of value creation. In tech industries, dynamic risk assessments help firms identify projects that offer competitive advantages while minimizing risks of technological obsolescence. Embedding such strategies enhances resilience and aligns projects with long-term goals. Industries like aerospace and telecommunications benefit most from integrating adaptive risk management and strategic alignment into PPM, driving innovation and sustainable success. This framework calls for further empirical research to refine PPM practices, fostering agility and strategic alignment across industries facing rapid technological and market changes.

Limitations and Future Study Suggestions

While offering valuable insights, this study acknowledges limitations including potential unconsidered factors that influence value generation and constraints related to sample size and the specific project management approaches used. Future research should explore the implications of identified criteria on project value and consider the impact of organizational culture on continuous learning and innovation. Expanding the scope to include various organizational sizes and project management methodologies could yield different results.

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Declarations

Conflict of interest The authors declare no conflict of interest.

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Key Questions

1. How can organizations effectively balance traditional financial metrics with non-financial criteria to evaluate project portfolio value and ensure strategic alignment?
2. In what ways can flexible project portfolio management (PPM) systems adapt to shifting stakeholder priorities and external market dynamics while maintaining alignment with long-term organizational goals?
3. How do risk and change management strategies contribute to enhancing portfolio value, and what best practices can be implemented to optimize their impact?
4. What are the implications of integrating advanced decision-support systems, such as fuzzy set theory, into PPM processes, and how can they improve strategic decision-making under uncertainty?

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